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1. INTRODUCTION

1.1 Background of the Study

Digital documents can be of many types and contain different kinds of information. Most organizations and educational institutions generate and retain these documents to keep track of records or other data about students, reports, and all of the administrative data that is part of running an organization or school. Frequently, the information will exist in many different formats due to having different needs for that information.

Since people most commonly convert their information between several formats while retaining original content, there is a great deal of this type of activity performed in an educational setting. For instance, some organizations that maintain student or educational operational information utilize Excel, MS Word, and PDF files which are also needed in alternate file formats for various purposes at various points in time (e.g., audits, inspections) to ensure that a method of verifying this data can be presented in a clear, concise manner.

It can be a bit tricky to convert documents from one format to another. Documents may be in a variety of different formats such as PDFs, Word documents, Excel documents, etc., with each format containing a different structure for the files. This creates many challenges when trying to deal with various document formats and get them converted into the required formats.

Thus, it has become necessary for us to have a system capable of effectively managing many different document formats and converting them into structured and standardised outputs. A system like this would streamline the management of documents as well as provide a more manageable way to use and validate data.**1.2**

Problem Statement

A lot of organizations and educational institutions use many different formats of documents such as PDFs to store large amounts of data. While data does not change from one format to another, there can be many times when data is required to be viewed in a different format depending on the individual's need.

At this time, the process of converting documents can only be done by hand. Users must take data out from PDFs or other documents, change that data to be in another format, and then put it back into a document in its new form. This is time-consuming especially if the user has to convert a large number of documents and cause errors at all points in converting between formats missed data, incorrect information being entered and inconsistent formatting. In addition, the user will have to continuously be concerned about how to keep their documents of similar formats or use standard formats when creating new documents out of them.

Using different formats of documents can make document management inefficient and challenging particularly in instances like when conducting an audit or verifying data. In addition to being inefficient and challenging, there are more complex layers to the process of handling multiple document formats and obtaining relevant data due to varied document types.

1.3 Existing System

Today, people are using different software applications to do most of their document processing, including spreadsheets and document editors. Both of these types of software allow us to develop, modify, and arrange our documents according to the type of information we need to create.

A popular approach when it comes to document processing is to use templates to create documents that have a consistent format. For example, many reports, as well as many other types of documents, are typically produced in similar ways, so using templates is a good solution.

Also, many programs are available to convert documents into different formats such as converting a PDF file to a Word or Excel file. The use of various format conversion programs improves the ability to process and organize your documents in different formats.

In short, the current systems provide a basic level of document processing and converting formats. It still requires users to manually extract data from documents and organize them.

1.4 Drawbacks of Existing System

Some of the drawbacks of existing document conversion and formatting systems include:

- Extraction and formatting processes take more time and effort if performed manually on large documents.
- Manual processes are highly prone to errors because of incorrect data entry, lost information, or dissimilar formats.
- Conversion tools are used to convert files to a different format but do not produce consistent outputs and require manual work to put into a consistent and structured format.

1.5 Smart Document Processing System

The proposed system provides automated methods to handle information from documents by converting them into structured formats. Users may provide input information through many different document types (e.g. PDF, Word, Excel).

The input information from users will be stored in a central database for retrieving and managing user input data. The retrieved information will then be converted into required format based on the user's need. The same information can be converted into multiple different structured types without the need for repeated manual conversion or processing.

The proposed system uses predefined models to ensure consistency and uniformity across outputs and creates an automated method for storing, retrieving, and converting information. Creating an automated method for managing data creates a reduction of the amount of manual labour required to perform document processing.

In comparison to existing systems, the proposed solution will show a more structured and dependable way to process documents while reducing errors and providing uniform output.

1.6 Objectives

- To collect a variety of documents used in institutions.
- To extract and store the data in a database.
- To convert the data into the required predefined format.

1.7 Methodology

The proposed system will an effective and organized way of processing documents and it is a structured approach. The methodology begins with the identifying the documents which are commonly used documents in educational institutions, such as question papers (IQAC), committee lists, and timetables. These documents are selected as modules because they are frequently used and produced in structured or semi-structured formats.

It enables users to upload documents of various formats, including PDF, Word, and Excel. Once the document is uploaded, it is processed according to the selected module. Each module has a unique purpose handle a specific type of document and extract relevant information from it.

The collected data must be cleaned before it is structured in pre-defined model formats. This guarantees similar outputs for consistent output. The user receives a preview of the processed data to verify before generating the final output.

After confirmation, the structured data will be generated in the desired format and stored in the database for future use. This solution minimizes the need for manual effort, increases the accuracy, and ensures efficient document processing.

1.8 Organization of the Project

The report includes different chapters related to the project.

Chapter 1: It consists of Introduction which includes background information, problem statement, currently existing system, proposed system, objectives, and methodology.

Chapter 2: It includes literature survey with details about other related works in terms of document processing.

Chapter 3: focuses on designing the system that includes system architecture and modules of the system.

Chapter 4: addresses how the system will be implemented with the technologies used and how each module will work.

Chapter 5: contains results and discussion of outputs, performance , and comparisons.

Chapter 6: Conclusion with future scope for the project.

2.LITERATURE SURVEY

Document processing has become a very common area of study for researchers in recent times because of the increased digitization of documents in the education sector . Document processing has mainly been done manually , where the user uses a spreadsheet application or word processor to extract information and organize data. The process can also be described as difficult and often prone to errors. As pointed out by Smith [13], semi-structured documents that have to be interpreted are inefficient when extracted manually.

In order to solve these problems, attempts have been made to automate the process of document processing, where unstructured and semi-structured documents are converted into structured documents. Automation of data extraction leads to increased efficiency through reduction of manual effort and faster data extraction. As discussed by Zhang [14], there are intelligent systems for document analysis that use structured extraction techniques to structure the contents of a document.

Template based approach is another commonly used technique when it comes to document processing. In this technique, the template is used to retrieve data from those documents having a certain structure. According to Katti and Arun [12], using templates is easy and quick and can be employed for analysing reports, extracting table data, and also developing record-keeping systems. It is useful in educational organizations where there are standard structures of reports such as IQAC, attendance list, and time table.

Data Extraction and classification is another important process within document processing. Data extraction is one of the areas that have been thoroughly researched using machine learning techniques. Some of the classification methods that can be used to extract data from documents include boosting & bagging algorithms [1] and support vector machines [4]. In addition, as pointed out by Goodfellow et al. [11], deep learning improves the performance of document analysis tools in terms of pattern recognition. However, data extraction methods may require lots of resources such as data and computer processing capability.

Besides machine learning, there are various other studies involving image processing and remote sensing that have contributed to automated data extraction. Some of these include the research done by Arif Janov et al. [2], Blanche et al. [3], and Drăguţ et al. [5]. The research focuses on segmentation and classification approaches used in pattern recognition of large amounts of data. Similarly, the research conducted by Dronova et al. [6], Duro et al. [7], Fassnacht et al. [8] stresses the significance of using automated techniques for data extraction and processing. These approaches are somewhat focused on image processing but could be modified to be used in document processing as well.

The aspect of data storage and management is equally significant in the context of document processing systems. Proper storage ensures that data is readily accessible, alterable and reusable. As highlighted by Lalita Parameswari et al. [9], structured data storage is required to ensure data integrity and reduce data redundancy. Database technology plays a critical role in data storage for its accessibility and management. The importance of structured storage systems for the management of voluminous data, as mentioned by Raja Sekharan and Pai [10], is equally relevant to this discussion. Web-based document processing systems have made these systems increasingly user-friendly. Through web-based document processing systems, individuals are able to upload their documents to process them in the cloud and save the results in the database. The process has become convenient through the use of technologies such as Python and Flask [15][16]. While Python is suitable for processing of data, Flask plays the role of client-server communication. Additionally, Pandas and OpenPyXL help these technologies interact with Excel data [17][18]. HTML and CSS technologies, amongst others, enhance the user experience as well as the appearance [19].

Despite all advancements in automation and machine learning approaches, there is a need for straightforward and effective solutions for structured/semi-structured document analysis. It is not always necessary to use advanced models for tasks related to the analysis of documents that have a pre-defined structure. In such cases, rule-based and template-based techniques would prove to be a better choice since they are faster, more efficient, and use lesser computing power [12].

From an analysis of relevant literature, it can be seen that automation and structured processing/storage are key elements in document processing. The machine learning techniques being applied here are sophisticated, but not always necessary for such tasks. A rule-based system supplemented by structured processing will suffice for this purpose.

This Smart Document Processing System is grounded on all the above findings. It aims at processing the documents that come under the academic domain, such as IQAC report documents, committees list and timetables. The system supports different methods of inputting information, then captures and organizes the data and finally stores it in the database for future use. Using the concepts of templates and web technology, the document processing process becomes easy and fast.

In summary, the literature reveals that automation, consistency, and data management are essential in the document processing system. The current system incorporates these concepts to address the limitations associated with the conventional systems.

3.SYSTEM OVERVIEW

3.1 Architecture of the System

The objective of the Smart Document Processing System is to allow automated extraction, structuring, and archiving of data present in multiple types of documents. The Smart Document Processing System is based on the concept of modularity, where each module in the system performs certain tasks.

The sequence of operations within the Smart Document Processing System includes uploading a document in the system using the interface that has been provided. Different forms of input can be used in the process, such as Excel, PDF, and Word documents. Upon uploading the document, it is stored temporarily and then forwarded to the processing stage.

The decision taken by the user directs the input into different modules like IQAC, Committee, and Timetable modules. The document is processed by each particular module based on the logic of that module.

Lastly, the system transmits the result of the process to the confirmation module, in which the user receives the preview of the processed data. Because each module uses different formats for previews, the preview will be executed depending on the used module. The purpose of doing this is to enable the user to verify if the result was processed properly.

As soon as the user approves the output, the system provides the user with the finalized output that appears in the pre-defined Excel format. In addition, the processed data is saved into the database for further use.

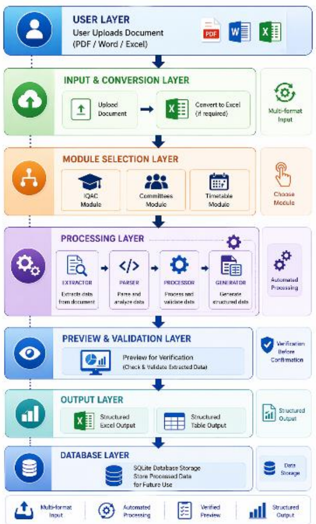


Fig 3.1. Architecture Diagram

3.2 Input Handling

The first process in the system entails receiving the data entry from the user interface through submission of the required document. After submission of the document, the system checks whether there has been submission of the right type of document. In case there is no submission of any document, then an error message will be generated.

In case the document uploaded is verified successfully, it is stored in a specified folder, and the file path is provided by the system for further processing.

Input Provided - Type of Document: Excel, PDF, or Word Document.

Output Produced - File Path.

The purpose of this step is to ensure that each document received is authentic and can go through subsequent processing stages.

3.3 IQAC Module

The IQAC stands for Internal Quality Assurance Committee. This is an initiative taken up by the educational institutions in order to maintain and enhance the quality of their education. The IQAC categorizes the question paper data according to the categories formed for questions based on COs (Course Outcomes) and BTs (Bloom’s Taxonomy Levels).

With the help of the IQAC Module, the information on the academic question paper available in the input document will be structured to form the final output. In the Input document, we have the number of questions, course outcomes, BT Levels, marks, and number of parts per question.

The first phase involved the extraction of data required from the Input document. However, since this data might not have been in proper order, therefore, the second phase involved the cleaning of data. In the phase of data cleaning, all of the data is standardized to ensure uniformity in the data format.

The data is then arranged according to the pre-established format and is exported as an Excel file. The data is also classified in suitable Course Outcomes (COs) or Bloom’s Taxonomy (BT).

Input: An IQAC Data Input Document

Output: An Excel file that contains IQAC Question Paper Data as well as the data contained in the database.

3.4 Committees Module

Objectives of the Committees module are designed to assist us to scan various kinds of tabulated documents such as data about project teams, and many more, which may appear in an educational institution environment. The documents are usually tabulated in form of rows and columns.

The ability of the module to process an input file and extract all the tabulated contents, whether they are rows or the headings, and put them into a table is something this module can do. This module can retain the tabular format of the extracted data and will make this tabular arrangement of data visible and available for viewing.

The module's input is any type of tabular document (Excel, PDF, Word) and produces an organized table of the extracted data. This module makes it easy for you to manage tabulated documents with the direct extraction and presentation of the documents as organized data.

3.5 Timetable Module

Timetable module aims to assist you in analysing different documents which have class timetable information and converting them into an organised form. These documents may be in the form of Excel, Word, or PDF documents.

The module will look at the following factors: the day of the week, time, subjects, teachers of those subjects, and any other related data. After that, the module will find patterns in the structure of the document. Upon analysis of the document, it will be converted into a table that can be easily utilized.

Input: Document timetable (in Excel, PDF, or Word form)

Output: A table timetable.

This module represents the first step towards automated data processing.

3.6 Data Storage

The information would be stored in the database after processing of data within various modules. This implies that the processed data will be stored in a secure manner to facilitate easy retrieval of data whenever needed.

These tables store the data in a database. There will be creation of a new table by each module for every document that has been processed. Therefore, we find it easy for data manipulation and retrieval.

Input: Processed structured data from modules

Output: Data stored in database tables

Data is stored in the database for future analysis purposes as well as to avoid processing the same data again.

3.7 Summary of System Flow

The entire process of this system has an easy flow. The user will first upload a document that will be validated and saved. Then the file will go through the respective module according to the choice of the user. Once the user chooses a module, it will convert the document into that particular structured form. In the end, the output is generated in excel and the processed data is stored in the database.

4.IMPLEMENTATION

4.1 Technologies Used

Deployment of Smart Document Processing System involves different types of backend technologies, data processing, and web development tools to ensure efficient document processing. Software architecture includes different document processing modules, data extraction techniques, and web-based interface. All of them are developed in such a way to ensure flexibility, efficiency, and maintainability. This chapter will focus on the use of technologies in the software system.

1. Python

The reason why the language Python is used to develop this particular software application is due to its user-friendly characteristics. Also, this particular language possesses those libraries which make it effective for file handling and automation purposes. This particular programming language would form the core of this project when it comes to document processing and module integration.

2. Flask

The Flask framework helps in building the backend of the application. The role of the framework is to handle routing, file upload, and communication between the front-end and back-end parts of the application. Flask makes it easier for the transfer of data between the UI components and processor modules, and thereby enables the user to do tasks such as file upload and selecting processor modules.

3. Pandas

Pandas library is used to process and analyze excel sheets within the system. The library has highly optimized data structures such as Data Frame that help in the importation, sorting, and manipulation of data. Pandas library can be efficiently used to cleanse and format extracted data.

4. OpenPyXL

OpenPyXL facilitates the process of creating and managing Excel files. This allows the system to create output files that are well-structured and adhere to the templates designed by the system. OpenPyXL will facilitate the process of data entry within Excel sheets.

5. Regular Expressions (Regex)

Regular expressions can be utilized in order to extract information based on certain patterns from the input document. These can prove useful when trying to identify some kind of structure in the document like question numbers, codes, etc. Regex has a vital role in structuring the information.

6. HTML and CSS

The HTML and CSS languages have been used in developing the user interface of the software. In the HTML language, web pages are developed for the user interface. In the CSS language, the web pages are styled. These languages provide an easy-to-use interface for file upload, module selection, and results display.

7. SQLite Database

The database management system adopted is SQLite, and the processed data will be saved into the system using this DBMS. SQLite is a light database management system that does not require any servers because it operates directly through the software itself. This is the best choice for small scale operations.

The system uses databases to store the acquired information. For instance, when it comes to Committee module, tables will be dynamically created based on the input documents. When it comes to the rest of the modules, the structured information will be saved in uniquely titled tables.

4.2 System Design

This system will be developed using a modular approach as a web application for processing documents using a pipeline-style methodology. This architecture comprises three main layers – the input layer, the processing layer, and the storage layer.

Input layer layer accepts file uploads from the user in different formats such as Excel, PDF, or Word files. The uploaded files are temporarily stored and are then sent to the processing layer.

Processing layer is at the heart of the entire system where the module chosen by the user is used to process the input document according to the module's unique algorithm.

Storage Layer stores the processed data will be stored in a database for future use, thus eliminating any need for further processing in the future.

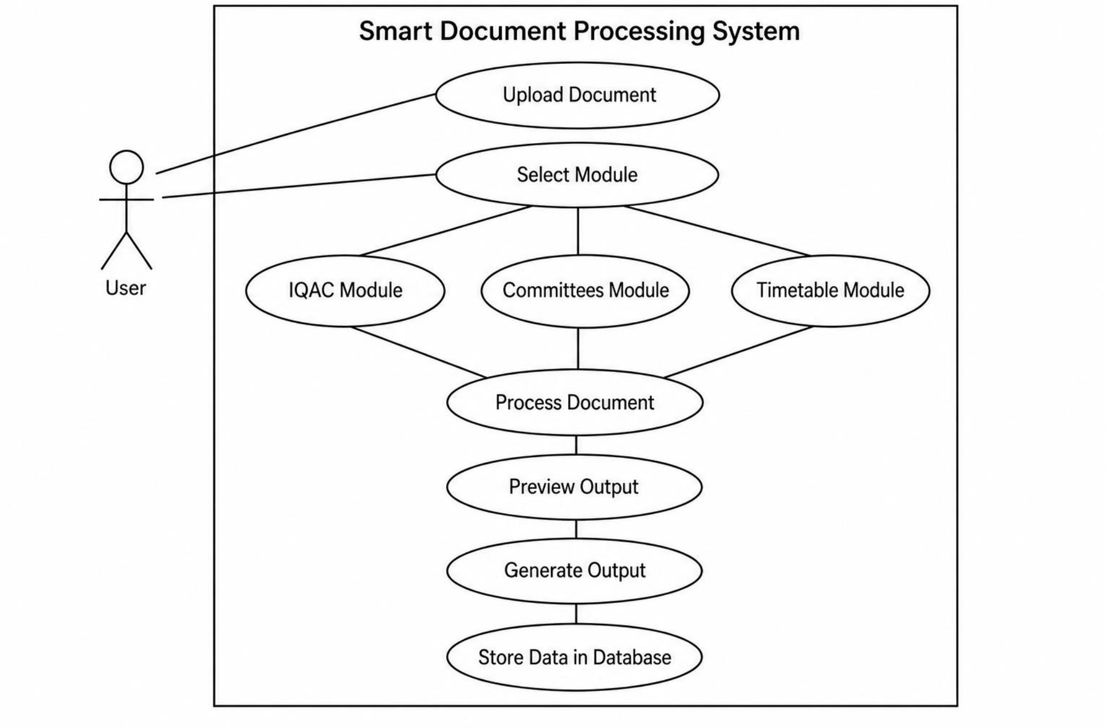


Fig 4.2. Use Case Diagram

4.3 Modules Implementation

4.3.1 IQAC Module Implementation

The purpose of the IQAC module is to arrange the data of question papers in an organized manner according to certain parameters of academic evaluation, which include Course Outcomes (COs) and levels of Bloom's Taxonomy.

The procedure used in the implementation of the IQAC module can be seen as a processing pipeline that includes several components like the extractor, parser, processor, generator, and service.

In the extractor, the input file is read and the required information such as question number, course outcome, Bloom's level, marks, and part is extracted. Inconsistencies are likely to exist in the extracted information. This is resolved in the parsing phase where inconsistencies and missing values are fixed. The processor is responsible for the entire process.

During the generator phase, the processed information will be mapped on the Excel spreadsheet. The questions are classified based on the Bloom's Taxonomy and the desired outcomes for each course. Service is the mediator that links the module to the core system.

4.3.2 Committees Module Implementation

This module addresses the issue of working with tabulated information, especially that contained within educational documents like committee listings and project teams. Tabulated information is mostly arranged in rows and columns.

The document is read by the system and processed as a tabular document, and later saved in the database as such. The document is read, and the title of the document, which serves as the table name, is obtained. This is followed by getting the column names and rows of data.

When the data is fragmented or is not fully captured due to splitting into rows, all these fragments of data are combined to form a single entity. Once processed, the data is saved as a table in the database.

4.3.3 Timetable Module Implementation

This module is capable of dealing with many different input document types like Excel, Word, and PDF documents. First of all, all the non-Excel documents are converted into a consistent Excel form.

Upon the analysis of this document, the program determines the data that concerns timetable items such as date, time slots, course name, and instructor name. The program detects patterns in the input document and pulls out the necessary elements.

Next, the extracted data is formatted into an easily accessible form. That is, it is organized in the form of a table in rows and columns that represent different days, times, courses, and instructors.

The data is cleaned and formatted further to ensure uniformity. The resultant data is presented in an easily readable tabular format. Finally, after confirming the results during the preview phase, the formatted timetable data is saved in a standard form in the database.

The module makes it easy for the user to manage the timetable, which is done through automation and minimization of errors. It also facilitates efficient handling of different file formats.

4.4 Data Storage Implementation

The processed data is kept in the database, which enhances its durability and accessibility. Each module keeps its output in the form of tables generated using the processed data.

For instance, the IQAC module generates the structured data which is kept in the database for later usage, whereas the Committees module forms the tables in the database itself.

Likewise, the Timetable module also generates structured data which is kept for later use. This enhances the efficiency in managing the data since there is no need to process the data again and again.

4.5 Workflow of the System

The process involves an orderly flow of operations. First, the user uploads a document via the user interface. This document is verified and then kept in temporary storage.

Thereafter, the user proceeds to choose a suitable module depending on the nature of the document. The chosen module performs an operation on the document according to its unique rules. Finally, the process produces structured information which is saved in the database.

User

↓

Upload Document

↓

Select Module

↓

Process Data

↓

Generate Structured Output

↓

Store in Database

5.RESULTS AND DISCUSSIONS

5.1 Results

The Smart Document Processing System successfully processes input documents such as IQAC reports, committee lists, and timetables. The system accepts multiple input formats including PDF, Word, and Excel, and converts them into structured outputs.

It generates organized data in the form of Excel files or tables and stores the processed information in the database. The system also provides a preview of the extracted data before final output generation, ensuring accuracy.

5.1.1 IQAC Module Results

The IQAC module was tested using question paper documents in PDF, Word, and Excel formats.

Initially, the user uploads the input document into the system, as shown in **Fig 5.1**. After uploading, the user selects the IQAC module, which is illustrated in **Fig 5.2**.

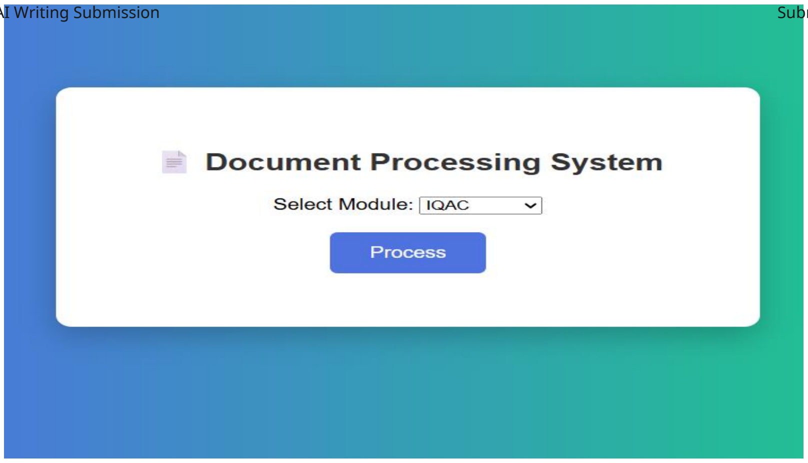
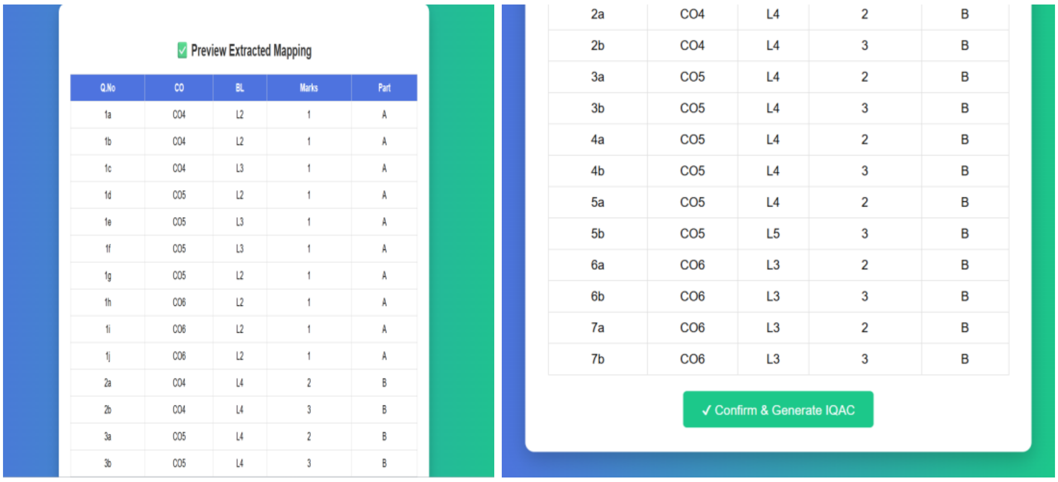


Fig 5.1 Uploading IQAC Input Document

Fig 5.2 Selecting IQAC Model

The system processes the document and generates a preview of the extracted data, allowing the user to verify the correctness of the information, as shown in **Fig 5.3**.



✅ Preview Extracted Mapping

Q.No	CO	BL	Marks	Part
1a	CO4	L2	1	A
1b	CO4	L2	1	A
1c	CO4	L3	1	A
1d	CO5	L2	1	A
1e	CO5	L3	1	A
1f	CO5	L3	1	A
1g	CO5	L2	1	A
1h	CO6	L2	1	A
1i	CO6	L2	1	A
1j	CO6	L2	1	A
2a	CO4	L4	2	B
2b	CO4	L4	3	B
3a	CO5	L4	2	B
3b	CO5	L4	3	B

2a	CO4	L4	2	B
2b	CO4	L4	3	B
3a	CO5	L4	2	B
3b	CO5	L4	3	B
4a	CO5	L4	2	B
4b	CO5	L4	3	B
5a	CO5	L4	2	B
5b	CO5	L5	3	B
6a	CO6	L3	2	B
6b	CO6	L3	3	B
7a	CO6	L3	2	B
7b	CO6	L3	3	B

✓ Confirm & Generate IQAC

Fig 5.3 Preview of IQAC Data

Upon confirmation, the system generates a

structured Excel output where questions are properly mapped based on Course Outcomes and Bloom's levels (**Fig 5.4**).

I. Distribution of marks for Blooms Taxonomy Levels:

S.No.	CRITERIA	Part-A		Part-B		0
			0		0	
1	Level1	1a,1b,1d,1	7		0	7
2	Level2	1c,1e,1f	3	6a,6b,7a,7	10	13
3	Level3		0	2a,2b,3a,3	17	17
4	Level4		0	5b	3	3
5	Level5		0		0	0
6	Level6					

II. Distribution of marks for Course outcomes

COURSEOUTCOMES	Part-A		Part-B		Total Marks
	Q.No.	Marks	Q.No.	Marks	
CO1		0		0	0
CO2		0		0	0
CO3		0		0	0
CO4	1a,1b,1c	3	2a,2b	5	8
CO5	1d,1e,1f,1g	4	3a,3b,4a,4	15	19
CO6	1h,1i,1j	3	6a,6b,7a,7	10	13

Fig 5.5 Generated IQAC Excel Output

Finally, the processed data is stored in the database for future use (**Fig 5.5**).

	q	co	level	marks	part
	Fil...	Filter	Filter	Filter	Filter
1	1a	CO4	L2	1	A
2	1b	CO4	L2	1	A
3	1c	CO4	L3	1	A
4	1d	CO5	L2	1	A
5	1e	CO5	L3	1	A
6	1f	CO5	L3	1	A
7	1g	CO5	L2	1	A
8	1h	CO6	L2	1	A
9	1i	CO6	L2	1	A
10	1j	CO6	L2	1	A
11	2a	CO4	L4	2	B
12	2b	CO4	L4	3	B
13	3a	CO5	L4	2	B
14	3b	CO5	L4	3	B
15	4a	CO5	L4	2	B
16	4b	CO5	L4	3	B
17	5a	CO5	L4	2	B
18	5b	CO5	L5	3	B
19	6a	CO6	L3	2	B
20	6b	CO6	L3	3	B
21	7a	CO6	L3	2	B
22	7b	CO6	L3	3	B

Fig 5.5 IQAC Data Stored in Database

This module successfully reduces manual effort and ensures consistency in IQAC report generation.

The IQAC module successfully processed input documents in multiple formats and generated structured Excel outputs. It accurately extracted question details and mapped them based on Course Outcomes and Bloom's levels. The preview and confirmation step ensured correctness before final output generation. Overall, the module reduced manual effort and improved consistency in IQAC report preparation.

5.1.2 Committees Module Results

The Committees module was tested using tabular documents such as committee lists and attendance sheets.

The user uploads the committees document into the system (**Fig 5.6**).

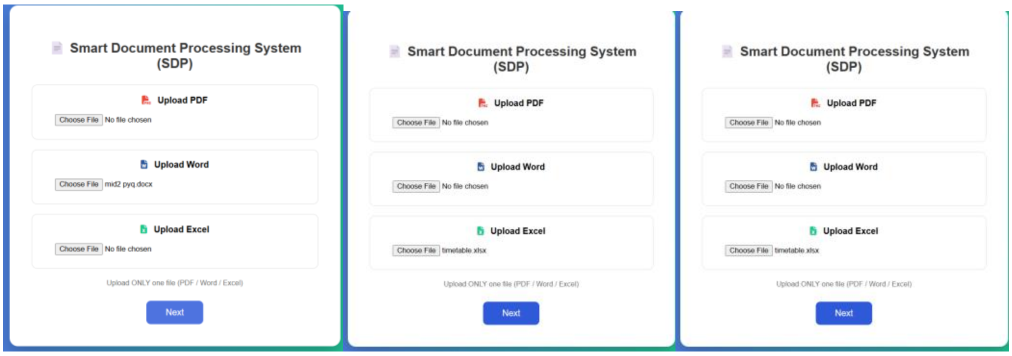
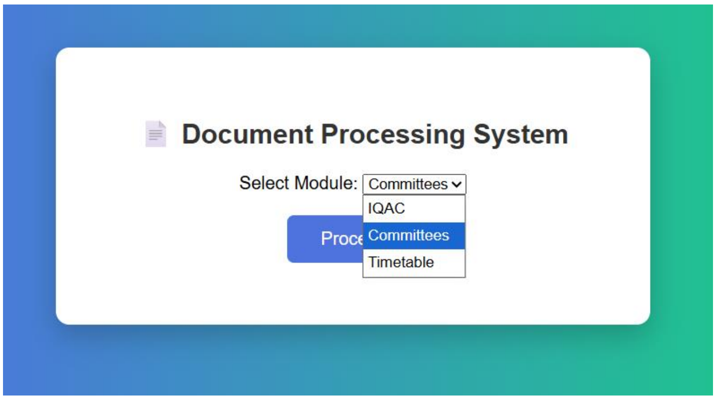


Fig 5.6 Uploading Committees Document

After uploading, the user selects the Committees module (**Fig 5.7**).

Fig 5.7 Module Selection – Committees



The system processes the document and displays a preview of the extracted tabular data (**Fig 5.8**).

Preview

S._No	Committee_Name	Required_No._of_Students_per Section	Dept._Faculty_Coordinator	Student_Coordinators_(II_B._Tech)
1	Computer Society of India (CSI)	2	Mrs. R. Mahatma	Mrudula-24251A05M7\nP.Hasini-24251A05R2
2	ISTE	4	Mrs. K. Sneha Reddy	Ramya sri-24251A05T0 Ch.Pranavi-24251A05L4 K.Samanvi-24251A05Q4 G.Sahitya-24251A05Q1
3	ACM-W	2\n(Ignore if\nalready there)	Mrs. Nandadevi D.R.	K.Parnitha-24251A05M1 P.Sreeja-24251A05N3
4	IEEE	2	Mrs. K. Sindhura	-
5	Placements	2	Mrs. Bhageshwari Ratkal Mr.Ch. Sudharshan Reddy Mr. G. Nagababu	G.Sahitya-24251A05Q1 Ramya sri-24251A05T0
6	College Magazine / News Letter	1	Mrs. Ch. Radhika	M.Sravanthi-24251A05Q8 T.Ruthika-24251A05R5
7	Wall Magazine (Chimera, AGAMA)	2	Mrs. B. Spandana	V.Sumedha-24251A05R8 M.Amulya-24251A05M6\nM.R.V sravanthi-24251A05Q9
8	Library Committee	2	Mrs. Nanda Devi. D.R	A.Supriya-25255A0522 S.Roshan-24251A05N7
9	Arts & Cultural Committee	2	Mrs. V. Divya Raj	A.sai sri Aruna-24251A05P1 B.Soumika-24251A05L1
10	NSS Committee	5	Mr. B. Vamshi	B.Divya-24251A05P6 P.Sai Priya-24251A05R3
11	Sports and Games Committee	2	Mr.Ch. Sudharshan Reddy	M.Siri Mahitha-24251A05Q6\nM.Anusha-24251A05M8
12	Innovation Cell	2	Mrs. M. Lalitha	P.Sreeja-24251A05N3
13	Hostel Committee	2	Mrs. Ch. Swathi	C.Harshitha- 24251A05R2\nM.Guna Priya-24251A05P0
14	Alumnae Association Committee	2	Mrs. J. Padmavathi	N.Saraswathi-24251A05N0\nK.sthitha pragna-24251A05M3
15	Career Guidance Cell	CR & ICR	Dr. P. Sunitha Devi/ Mr. P. Kranthi Kiran	Shruthi-24251A05N5\nP.Sri Hamsika-24251A05R4
16	NPTEL/FDP Committee	1	Mrs. Ch. Swathi	Guna priya- 24251A05P0
17	Press, Social Media & Publicity Committee	2 from all 5 sections	Mrs. G. Sandhya	Pari chopdar- 24251A05N4K. Parnitha-24251A05M1
18	Research Advisory Committee / R & D Cell	1	Dr. D.V. Lalitha Parameswari	T.Ruthika- 24251A05R5
19	Internal Complaints Committee	1	Mrs. Bhageshwari Ratkal	Sahana- 24251A05L7
20	Class Review Committee	6 + CR & ICR\n(1 lateral entry is must)	Mrs.J. Srilatha	Shruthi-24251A05N5\nP.Sri Hamsika-24251A05R4 K.Samanvi-24251A05Q4\nHema Harshini-24251A05L6 G.Guna Priya-24251A05P0 G.Sahithya-24251A05Q1 T.Nandika-24251A05R6\nCh.Rupa-25255A0525
21	Anti-Ragging Committee	(Already taken)	Mrs. R. Mamatha	-
22	Transport Committee	1 or 2	Mrs. A. Leela Kumari	P.Sai Priya-24251A05R3
23	Canteen Committee	1	Mr. B. Vamshi	P.Sai Priya-24251A05R3
24	Environmental Club	(Already taken)	Mr. B. Vamshi	-

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Fig 5.8 Preview of Committees Data

After confirmation, the system stores the processed data in the database as a structured table (**Fig 5.9**).

	A	B	C	D	E
1	S._No	Committe	Required_	Dept._Fac	Student_Coordinators_(II_B._Tech)
2	1	Computer	2	Mrs. R. Ma	Mrudula-24251A05M7P.Hasini-24251A05R2
3	2	ISTE	4	Mrs. K. Sne	Ramya sri-24251A05T0 Ch.Pranavi-24251A05L4 K.Samanvi-24251A05Q4 G.Sahitya-24251A05Q1
4	3	ACM-W	2(Ignore if	Mrs. Nand	K.Parnitha-24251A05M1 P.Sreeja-24251A05N3
5	4	IEEE	2	Mrs. K. Sin	-
6	5	Placement	2	Mrs. Bhag	G.Sahitya-24251A05Q1 Ramya sri-24251A05T0
7	6	College Ma	1	Mrs. Ch. R	M.Sravanthi-24251A05Q8 T.Ruthika-24251A05R5
8	7	Wall Maga	2	Mrs. B. Spa	V.Sumedha-24251A05R8 M.Amulya-24251A05M6M.R.V sravanthi-24251A05Q9
9	8	Library Co	2	Mrs. Nand	A.Supriya-25255A0522 S.Roshan-24251A05N7
10	9	Arts & Cult	2	Mrs. V. Div	A.sai sri Aruna-24251A05P1 B.Soumika-24251A05L1
11	10	NSS Comm	5	Mr. B. Van	B.Divya-24251A05P6 P.Sai Priya-24251A05R3
12	11	Sports and	2	Mr.Ch. Suc	M.Siri Mahitha-24251A05Q6M.Anusha-24251A05M8
13	12	Innovation	2	Mrs. M. La	P.Sreeja-24251A05N3
14	13	Hostel Cor	2	Mrs. Ch. S	C.Harshitha- 24251A05R2M.Guna Priya-24251A05P0
15	14	Alumnae A	2	Mrs. J. Pa	N.Saraswathi-24251A05N0K.sthitha pragna-24251A05M3
16	15	Career Gui	CR & ICR	Dr. P. Suni	Shruthi-24251A05N5P.Sri Hamsika- 24251A05R4
17	16	NPTEL/FDI	1	Mrs. Ch. S	Guna priya- 24251A05P0
18	17	Press, Soci	2 from all	Mrs. G. Sa	Pari chopdar- 24251A05N4K. Parnitha-24251A05M1
19	18	Research A	1	Dr. D.V. La	T.Ruthika- 24251A05R5
20	19	Internal Cc	1	Mrs. Bhag	Sahana- 24251A05L7
21	20	Class Revie	6 + CR & IC	Mrs.J. Srila	Shruthi-24251A05N5P.Sri Hamsika- 24251A05R4 K.Samanvi-24251A05Q4Hema Harshini- 24251A05L6 G.Guna Priya- 24251A05P0 G.Sahithya-
22	21	Anti-Raggi	(Already ta	Mrs. R. Ma	-
23	22	Transport	1 or 2	Mrs. A. Lee	P.Sai Priya-24251A05R3
24	23	Canteen C	1	Mr. B. Van	P.Sai Priya-24251A05R3
25	24	Environme	(Already ta	Mr. B. Van	-
26					
27					
28					

Fig 5.9 Committees Data Stored in Database

In the Committees module, the system successfully extracted tabular data from input documents and stored it in the database as structured tables. The system handled irregular data formats by cleaning and organizing the data before storage.

The Committees module effectively processed tabular documents and stored the extracted data in the database as structured tables. It handled irregular and incomplete data by cleaning and organizing it into proper formats. The module simplified the management of committee-related data and eliminated the need for manual data entry.

5.1.3 Timetable Module Results

The Timetable module was tested using timetable documents in different formats.

The user uploads the timetable document into the system (**Fig 5.10**).

Fig 5.10 Uploading Timetable Document

After uploading, the user selects the Timetable module (**Fig 5.11**).

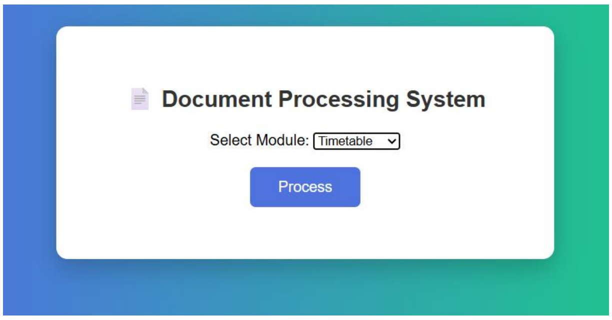

Fig 5.11 Module Selection – Timetable

The system processes the document and extracts timetable details such as day, time slots, subjects, and faculty information, which are displayed in preview format (**Fig 5.12**).

Timetable Preview

Structured Output

day	time	subject	faculty
Monday	09:00-10:00	COA	Mr.Ch.Sudharshan Reddy
Monday	10:00-11:00	OOPJ	Dr.T.Divya Kumari
Monday	11:10-12:10	HVPE	Mrs.Durga Bhavani
Monday	12:10-1:00	LUNCH	
Monday	1:00-4:00	OOPJ Lab/OSMP LAB	Dr.T.Divya Kumari / Mr.Ch.Sudharshan Reddy / Mrs.Ch.Radhika / Mrs.R.Mamatha
Tuesday	09:00-12:10	Placement Training	
Tuesday	12:10-1:00	LUNCH	
Tuesday	1:00-4:00	Placement Training	
Wednesday	09:00-10:00	OOPJ	Dr.T.Divya Kumari
Wednesday	10:00-11:00	DM	Mrs.R.Pallavi
Wednesday	11:10-12:10	SE	Mrs.Subha Lakshmi
Wednesday	12:10-1:00	LUNCH	
Wednesday	1:00-2:00	COA	Mr.Ch.Sudharshan Reddy
Wednesday	2:00-4:00	MP-1	Dr.T.Divya Kumari / Mrs.V.Subha Lakshmi
Thursday	09:00-10:00	OS	Mrs.R.Mamatha
Thursday	10:00-11:00	DM	Mrs.R.Pallavi
Thursday	11:10-12:10	OOPJ	Dr.T.Divya Kumari
Thursday	12:10-1:00	LUNCH	
Thursday	1:00-2:00	HVPE	Mrs.Durga Bhavani
Thursday	2:00-4:00	MP-1	Dr.T.Divya Kumari / Mrs.V.Subha Lakshmi
Friday	09:00-12:10	OSMP Lab/OOPJ Lab	Dr.T.Divya Kumari / Mr.Ch.Sudharshan Reddy / Mrs.Ch.Radhika / Mrs.R.Mamatha
Friday	12:10-1:00	LUNCH	
Friday	1:00-2:00	OS	Mrs.R.Mamatha
Friday	2:00-3:00	SE	Mrs.Subha Lakshmi
Friday	3:00-4:00	Library/Sports	
Saturday	09:00-10:00	SE	Mrs.Subha Lakshmi
Saturday	10:00-11:00	COA	Mr.Ch.Sudharshan Reddy
Saturday	11:10-12:10	OS	Mrs.R.Mamatha
Saturday	12:10-1:00	LUNCH	
Saturday	1:00-2:00	DM	Mrs.R.Pallavi
Saturday	2:00-4:00	Department Activities	

✓ Confirm & Download

Fig 5.12 Preview of Timetable Data

After confirmation, the system generates structured timetable data and stores it in the database (**Fig 5.13**).

	day	time	subject	faculty
	Filter	Filter	Filter	Filter
1	Monday	09:00-10:00	COA	Mr.Ch.Sudharshan Reddy
2	Monday	10:00-11:00	OOPJ	Dr.T.Divya Kumari
3	Monday	11:10-12:10	HVPE	Mrs.Durga Bhavani
4	Monday	12:10-1:00	LUNCH	
5	Monday	1:00-4:00	OOPJ Lab/OSMP LAB	Dr.T.Divya Kumari / Mr.Ch.Sudharshan...
6	Tuesday	09:00-12:10	Placement Training	
7	Tuesday	12:10-1:00	LUNCH	
8	Tuesday	1:00-4:00	Placement Training	
9	Wednesday	09:00-10:00	OOPJ	Dr.T.Divya Kumari
10	Wednesday	10:00-11:00	DM	Mrs.R.Pallavi
11	Wednesday	11:10-12:10	SE	Mrs.Subha Lakshmi
12	Wednesday	12:10-1:00	LUNCH	
13	Wednesday	1:00-2:00	COA	Mr.Ch.Sudharshan Reddy
14	Wednesday	2:00-4:00	MP-1	Dr.T.Divya Kumari / Mrs.V.Subha ...
15	Thursday	09:00-10:00	OS	Mrs.R.Mamatha
16	Thursday	10:00-11:00	DM	Mrs.R.Pallavi
17	Thursday	11:10-12:10	OOPJ	Dr.T.Divya Kumari
18	Thursday	12:10-1:00	LUNCH	
19	Thursday	1:00-2:00	HVPE	Mrs.Durga Bhavani
20	Thursday	2:00-4:00	MP-1	Dr.T.Divya Kumari / Mrs.V.Subha ...
21	Friday	09:00-12:10	OSMP Lab/OOPJ Lab	Dr.T.Divya Kumari / Mr.Ch.Sudharshan...
22	Friday	12:10-1:00	LUNCH	
23	Friday	1:00-2:00	OS	Mrs.R.Mamatha
24	Friday	2:00-3:00	SE	Mrs.Subha Lakshmi
25	Friday	3:00-4:00	Library/Sports	
26	Saturday	09:00-10:00	SE	Mrs.Subha Lakshmi
27	Saturday	10:00-11:00	COA	Mr.Ch.Sudharshan Reddy
28	Saturday	11:10-12:10	OS	Mrs.R.Mamatha
29	Saturday	12:10-1:00	LUNCH	
30	Saturday	1:00-2:00	DM	Mrs.R.Pallavi
31	Saturday	2:00-4:00	Department Activities	

Fig 5.13 Timetable Data Stored in Database

For the Timetable module, the system extracted relevant details such as day, time, subject, and faculty information, and converted them into a structured tabular format. The output was clear and easy to interpret.

The Timetable module successfully extracted timetable details such as day, time slots, subjects, and faculty information from input documents. It converted the data into a structured and easy-to-understand format. The module ensured clarity and accuracy in timetable representation, making it suitable for further use and analysis.

Overall, the system successfully processed documents across all modules and generated accurate and structured outputs. It reduced manual effort, minimized errors, and improved efficiency in handling real-world documents.

5.2 Performance Analysis

Evaluation criteria for this system included efficiency, effectiveness, ease of use, and flexibility. The system demonstrated efficiency by cutting down processing time of the document. Activities such as manually reviewing, editing, and formatting have been automated and performed within a very short time.

Precision was also achieved since the system reduces any human errors in the process of extracting and organizing information. The use of structured processing means that the output will always be standardized regardless of the document. Moreover, with the preview tool, one can confirm the information before generating the output.

It is very easy to work with this software because it offers a user-friendly interface where the documents are uploaded and the output is displayed without any technical expertise needed by the users. The flexibility in the software arises because of the multiple formats of the inputs that can be fed into the system, such as PDF, Word, and Excel formats.

Overall, the system improves efficiency, reduces errors, and provides a simple and effective solution for document processing.

The table below compares the existing system with the proposed system (see Table 5.2.1).

Table 5.1. Comparison of Existing System vs Smart Document Processing (SDP)

Feature	Existing Systems	Proposed System
Document Processing	Manual processing using Excel/Word	Automated document processing
Data Extraction	Manual extraction	Automatic extraction
Processing Time	Time-consuming	Fast processing
Accuracy	Error-prone	High accuracy
Database Storage	Not used	Stored in SQLite database

5.3 Discussion

This created system will become an effective approach for overcoming issues related to traditional documentation management. At present, technologies make use of Microsoft Office packages like Excel and Word, which require previous experience and much effort. This is not only time-consuming but also causes errors.

Unlike existing approaches, the suggested approach will be able to carry out all these actions without any manual intervention: extracting, structuring, and even placing information into a database. This will make the process more effective because the importance of manual work will be diminished.

It becomes quite significant to educational institutes that have regular requirements of processing IQAC reports, timetables, committee reports, etc. The automation of all these tasks makes the job easier for them.

6.CONCLUSIONS AND FUTURE ENHANCEMENTS

6.1 Conclusion

This system was invented and developed with the objective of automating the varied activities involved in processing regular academic-related documents. The SDPS processes IQAC reports, committee logs, and timetables to produce a well-organized form of data from the unorganized forms of data.

With automation, there will be less effort required to perform functions such as data extraction, data cleaning, formatting, and data storage. As a result, there will be minimal errors because of human involvement and, on the other hand, consistency will always be guaranteed. The fact that a large number of input file types, including Excel, PDF, and Word documents, are supported will make everything easier.

This is because there is a lot of flexibility involved in the designing of the system, considering that the components of the system can be operated alone. In addition, the inclusion of the database will enhance the performance of the entire system.

6.2 Future Scope

Although the system works efficiently, there are certain areas that could be improved. They are adding support for different file formats and supporting complicated documents. This would increase the flexibility of the system and could be used in more applications.

In order to increase the reliability of information collected from documents, the application of modern technologies, such as machine learning and natural language processing, should be considered.

Also, the system can be enhanced through the integration of additional cloud services that provide both remote accessibility and enhanced data management capabilities. Analytics and reporting features could be added to improve the analytic insight while accessing the data.

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GLOSARRY

1. **Smart Document Processing (SDP)**

Definition: SDP is an automated tool used for retrieving, processing, and structuring information from unstructured documents.

2. **IQAC (Internal Quality Assurance Cell)**

Definition: Education-based structure used for maintaining and enhancing academic quality.

3. **Module**

Definition: A subsystem of the overall system that performs one function, such as IQAC processing, extracting timetables, or handling committee data.

4. **Extractor**

Definition: A module that accepts input files and extracts necessary information.

5. Parser

Definition: A component for cleaning and normalizing the data obtained from other sources into a consistent format.

6. Processor

Definition: A component that manages the process of how tasks are to be executed within the overall data processing cycle.

7. Generator

Definition: A part that takes care of translating the information into a certain format like the Excel sheet.

8. Service Layer

Definition: An element that integrates the components and handles the communication between them and the application.

9. Flask

Definition: A basic framework for web development in Python to develop the application back-end.

10. SQLite

Definition: A lightweight and file-based DBMS to keep the processed data stored on a local machine without the need for any external server.

11. Database

Definition: A database is the systematic way of keeping the processed data for use in the future.

12. Table

Definition: Table refers to data present in a database that have been arranged in rows and columns.

13. Extracting Data

Definition: To extract useful data from the documents being used as inputs.

14. Data Processing

Definition: It involves the process of cleaning, organizing, and transforming the extracted data into a structured format.

15. A Structured Data Definition:

Definition: Helps in storing and analyzing data with ease.

16.Template

Definition: A template refers to the pre-set format used to produce standardized output documents.

17. Preview

Definition: Preview refers to a facility that allows users to view data after processing it prior to the generation of output.

18.User Interface(UI)

Definition: UI refers to the interface that depicts how a user interfaces with the computer.

19. Data Validation

Definition: Data validation refers to the essential process of validating extracted data to ensure that it is correct and complete before generating output.

20. Workflow

Definition: Work flow is the order in which the user performs the steps from the moment they upload their input file until they produce their output and save the data in the database.

21. Automation

Definition: Automation is the use of the system's logic to execute certain actions automatically without any user intervention.

22. File Conversion

Definition: File Conversion is the process of converting the file format of the input file (PDF/Word) to an Excel format file in order for it to be processed.

23. Back-end

Definition: Back-end is the server-side part of the system that does all the processing and handling of the data.